HELLO WORLD!

MD ASIF

Propulsion & Vehicle Dynamics Engineer specializing in structural FEA analysis, multi-axis kinematic systems, and Stirling thermodynamic cryocooler simulations.

EXPLORE PROJECTS ↓ VIEW CAREER HISTORY

THERMODYNAMICS ✦ FEA & CFD ANALYSIS ✦ KINEMATICS ✦ VEHICLE DYNA
ANICAL ✦ FLUENT CFD ✦ MATLAB / PYTHON

FEATURED CASE STUDIES

SELECTED PROJECTS

01

WHEEL ASSEMBLY

VEHICLE DYNAMICS / FEA / MANUFACTURE

PROJECT SUMMARY

- Designed custom lightweight hub & steering knuckle for ESVC solar car.
- Optimized corner structural integrity using Finite Element Analysis (FEA).
- Reduced total unsprung mass while preserving critical load handling parameters.

PROJECT PDF

CAD MODEL • STEERING KNUCKLE ASSEMBLY

Wheel Assembly Steering Knuckle CAD

ESVC SOLAR CAR HUB / FEA PASS

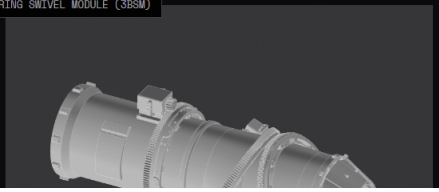
02

F35-3BSM KINEMATICS

REVERSE ENGINEERING / CAD / PRODUCT DESIGN

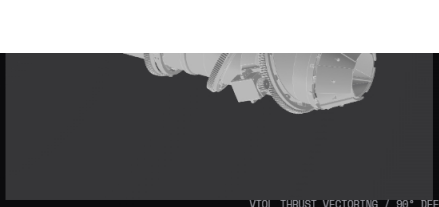
PROJECT SUMMARY

3-BEARING SWIVEL MODULE (3BSM)



– Reverse-engineering the F-35 thrust-vectoring nozzle kinematics layout.
– Simulating multi-axis 3-bearing gear rotation matrices for VTOL transitions.

PROJECT PDF



VTOL THRUST VECTORING / 90° DEFLECTION

0 3

STIRLING CRYOCOOLER

THERMODYNAMICS / FEA & CFD / CAD

PROJECT SUMMARY

– Formulated 1D thermodynamic model in Python/MATLAB via Schmidt theory (5 W at 80 K).
– Engineered 3D CAD assemblies for compressor, mesh regenerator, and displacer in SolidWorks.
– Analyzed CFD/transient thermal mesh matrices and simulated COP to yield 15% efficiency boost.

WORK IN PROGRESS

Schmidt Theory 1D Matrix • 5W at 80K Thermal Regenerator Mesh Simulation

COP SIMULATION: +15% EFFICIENT

PROJECT PDF

• WORK IN PROGRESS

CAREER HISTORY & RESEARCH

EXPERIENCE

TEAM DTU SUPERMILEAGE

FULL-TIME • 1 YR 3 MOS
DELHI, INDIA • ON-SITE

VEHICLE DYNAMICS | LEAD

AUG 2025 - PRESENT • 10 MOS

Leading the Vehicle Dynamics division for DTU's high-efficiency prototype vehicles. Directing motor/engine integration, steering assembly design, suspension layout architectures, and cross-functional engineering management.

SOLIDWORKS | ANSYS MECHANICAL

PROPULSION AND DYNAMICS | CO-HEAD

AUG 2025

Co-directed initial team setup, vehicle dynamic simulation models, and subsystem layout integrations for prototype testing.

PROPULSION & DYNAMICS | MEMBER

MAR 2025 - AUG 2025 • 6 MOS

DRDO

DEFENCE RESEARCH & DEVELOPMENT ORGANISATION

RESEARCH INTERN • INTERNSHIP
DEC 2025 - JAN 2026 • 2 MOS

COMPLETION CERTIFICATE

DIRECT CONTACT & QUOTES

SEND THE DATA. GET A QUOTE.

WHATSAPP

TEXT ME

EMAIL

MAIL ME

LOCATION: NEW DELHI, INDIA

PORTFOLIO: MDASIF.IN

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